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Applying machine learning and natural language processing to detect phishing email

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ABSTRACT

The growth of online services has been accompanied by increased growth in cyber-attacks. One of the most common effective attacks is phishing, in which attempts are made to steal confidential information by impersonating a legitimate source. The success of phishing emails is based on manipulating human emotions, which leads to concerns and creates an urgent situation by claiming that the recipient should take quick action that may cause great financial and data losses. Therefore, we cannot rely solely on humans to detect phishing, and more effective and automatic phishing detection mechanisms are required. Many detectors have been proposed; however, the high number of phishing emails urges additional effort. Hence, in this study, we propose a phishing email classifier model that applies deep learning algorithms using a graph convolutional network (GCN) and natural language processing over an email body text to improve phishing detection accuracy. The literature has proved GCN success in text classification, and this study proved its success in improving the accuracy of email phishing detection. The classifier was tested in a supervised learning approach. Experimental tests verified that the classifier was effective in detecting phishing emails using body text among the existing detection methods, and it took short time and produced a high accuracy rate of 98.2% and a low false-positive rate of 0.015.

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1. Introduction

The rapid development of online services and the Internet has been regrettably accompanied by growth in cyber-attacks, with phishing being one of the most common and effective attacks (Halgaš, Agrafiotis and Nurse, 2020). Phishing is a type of cyber-attack that uses social engineering and technical tricks to steal a user's sensitive data (Khonji, Iraqi and Jones, 2013; Peng, Harris and Sawa, 2018a; Fang *et al.*, 2019; Halgaš, Agrafiotis and Nurse, 2020). It is done by exploiting vulnerabilities found in system processes from an end user's side; combating it depends on human knowledge and awareness, which can vary between users. Even if the system is secure enough against cybercrime, an oblivious user can be duped by an attacker into revealing their personal identity data or financial account credentials. Even worse, it can put an organisation at risk by facilitating access to the organisation's data. The global annual losses of cybercrimes calculated by McAfee are estimated to reach one trillion US dollars (McAfee, 2020). Moreover, phishing is a significant cybercrime that affects businesses and individuals (Sadique *et al.*, 2020).

According to a published phishing report, 75% of organisations experienced a phishing attack in 2020, and 96% of the phishing attacks arrived by email (Phishing Statistics Report, 2021). Of all the security breaches, 22% were related to phishing, according to Verizon's 2020 Data Breach Investigations Report (DBIR, 2020). Organisations have increasingly

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become targets for attackers attempting to steal money or, worse, their valuable data. The consequences of data theft are severe: it damages the organisation's reputation and value. Moreover, it causes customer loss for sectors that hold sensitive data, such as the banking and telecom sectors. Therefore, businesses should adopt a robust phishing detection mechanism.

The most commonly used phishing technique is phishing emails (Gupta, Arachchilage and Psannis, 2018; Phishing Statistics Report, 2021), by which an attacker uses a fake email to trick the recipient into sending their sensitive information. There are many methods to persuade a recipient to reveal their sensitive information, such as clicking a link and visiting a spoofed site to enter their username and password, sending information directly, or downloading an email attachment that executes malware. The increasing number of phishing emails is alarming, according to the Anti-Phishing Working Group report; the average cost for attacks grew in the third quarter of 2020 for businesses from $48,000 to $75,000 (APWG, 2020). Moreover, the number of recorded phishing emails globally in the fourth quarter of 2020 increased by 18% over the second quarter of 2019 (APWG, 2020). Furthermore, in 2020, the number of malicious emails in Microsoft 365 reached 13 billion messages (Microsoft digital defense report, 2020).

Benenson et al. (Benenson, Gassmann and Landwirth, 2017) reported a phishing experiment that showed that 20% of the audience responded and clicked the emails' specious link. When asked why they clicked the link, 34% of the respondents gave the reason that they were curious. They recommended that organisations protect employees, as much as possible, from the decision-making process of opening and responding to phishing emails. Consequently, the automation of phishing email detection using body email text is necessary to combat this growing risk.

Currently, we have various successful phishing email detectors, but businesses and individuals still suffer from phishing. The nature of phishing emails is dynamic because it depends on the human intelligence of an attacker. Over time, attackers further develop and change their phishing methods. Thus, we should confront this problem by finding a solution that matches human intelligence.

To deal with this problem, novel methods such as deep learning, which can compete with human intelligence, have been introduced, but these methods are still immature and need to be improved (Akanbi, Amiri and Fazeldehkordi, 2014). Furthermore, most of the proposed automated phishing email detection techniques to decrease human error use email metadata or attached URLs.

Previous works on phishing email detection have shown encouraging results, but more work on detection models is needed because the dynamic nature of phishing emails requires advanced methods. Again, few promising phishing detection models based on body text analysis have been proposed (Gupta, Arachchilage and Psannis, 2018), and more effort is needed. In addition, the graph convolutional network (GCN) algorithm has proved its efficiency in text classification (Yao, Mao and Luo., 2019), but no phishing email classifier based on body text analysis using a GCN has yet been proposed. This study is among the first, to the best of our knowledge, to investigate the effectiveness of using a GCN to detect phishing in email body text to improve phishing detection accuracy and speed. Moreover, the proposed classifier can detect phishing from emails even if the email has no URL or attachments.

In this study, we used email body text to enhance phishing detection accuracy, as we believe that there is a significant amount of information under the email body's text that is worth the effort. Consequently, we propose a new classifier for phishing email detection based on natural language processing (NLP), deep learning algorithms, and a GCN. The classifier determines whether an email is legitimate or a phishing attempt. To do this, the email body is analysed using NLP and deep learning algorithms. An evaluation of the accuracy of the classifier is then conducted using the metrics of accuracy, precision, and recall and compared with state-of-the-art models. On a balanced and labeled dataset, the classifier proposed in this research performs well. The accuracy exceeds 98%, and all evaluation metrics are superior to existing detection models.

The classifier investigation is only applied to English language emails and does not cover phishing websites or emails in other languages. This study does not include any analysis of email attachments or URLs. The proposed solution was implemented on the mail server to detect phishing emails. Nevertheless, it could have other applications in detecting phishing through text analysis in different common phishing mediums, such as social media messages, posts, or tweets.

The rest of this paper is organised as follows: Section 2 presents a research background that summarises some work on phishing email detection. Section 3 describes the proposed classifier model. Then, the classifier is tested and evaluated in Section 4. Finally, the conclusion is presented with an overview of the project's future work.

2. Research background

Phishing email is one of the Internet's fastest-growing cybercrimes. The goal of a phishing email is to steal the confidential information of a user by pretending to be a legitimate source. For cybercriminals, phishing has turned into a feasible business. Successful phishing attacks can cause financial loss for victims and put their personal information at risk. It affects even emerging technologies, including the Internet of Things and cloud services (Alieyan *et al.*, 2019; Al-Qerem *et al.*, 2020).

There are two main types of technical methods to detect phishing emails: blacklisting and machine learning. The blacklisting method compares the sender's email address, IP address, or DNS address with a predefined list of phishing addresses, and if the data match, the email is rejected before it reaches the SMTP mail server (Khonji, Iraqi and Jones, 2013; Gupta, Arachchilage and Psannis, 2018). While the blacklisting method has a low false-positive rate, it primarily depends on the reporting of phishing emails from the recipient side (Fang *et al.*, 2019). In the same vein, automatic whitelisting monitors incoming and outgoing emails and produces a collection of legitimate addresses. It also reviews records of past email communications stored within an email server for addresses to add to the list. Although whitelisting can be used to prevent phishing email, it is not effective enough to detect all phishing attacks (Jain and Gupta, 2019). It does not provide security

against zero-day attacks, as new addresses or sites cannot be detected.

Alternatively, machine learning can automate phishing email detection through various methods, such as deep learning detectors that automate phishing email detection. To improve phishing detection, researchers have proposed two methods to improve the classifier: 1) testing different algorithms, or 2) focusing on feature selection to improve the classifier (Toolan and Carthy, 2010).

Feature selection is defined as a method for obtaining a subgroup from an original feature group that matches specific feature selection measures. Through this process, the useful features of the dataset are identified. It helps reduce the data-processing size by removing unnecessary and irrelevant features. Good feature selection can increase machine learning accuracy (Jain and Gupta, 2019), simplify learning results, and decrease learning time (Cai *et al.*, 2018). Three main feature classes can be used in phishing email detection, as described in (Aassal *et al.*, 2020): *Header feature class,* which includes a timestamp, recipient server, content format, and other header information; *body feature class,* which includes two sub-classes (text features and URL classes); and *attachment class*.

Several studies prove the efficiency of using textual features of the text to enhance the performance of phishing classifiers (Jain and Gupta, 2019). Toolan and Carty (2010) studied 40 features used repeatedly in the literature. The features were categorised into four groups: body-based, URL-based, subject-based, and script-based features. In their study, the information gain for each feature was calculated, and then the classifiers for each feature were generated and evaluated. The results of this study were favourable for body-based features.

Bergholz et al. (2008) addressed advanced features for phishing detection. The authors noted the statistical insignificance of improving detection by modifying the classification algorithm itself and concluded that using enhanced features improved email phishing classification statistically significantly. Based on an unsupervised algorithm, two groups of advanced features were developed to complement 27 commonly used features in phishing detection. The basic features included structural features, link features, element features, word list features, and spam features. The advanced features included the dynamic Markov chain model, latent topic model features, and word clusters of topics based on latent Dirichlet allocation.

Moreover, machine learning techniques are commonly used to solve classification problems and detect phishing using deep learning algorithms or other algorithms such as naive Bayes and support vector machines SVM (Kubat, 1992). Deep learning algorithms have produced impressive results with unstructured data such as email data (Halgaš, Agrafiotis and Nurse, 2020).

Deep learning is one of the machine learning classes that uses algorithms to explore and model associated data, where a computer model can learn to execute a classification job directly using neural networks and datasets that include texts, sounds, or images. In this study, the reason for choosing deep learning over other machine learning algorithms is the extensive work of feature engineering used to implement conventional machine learning algorithms (Nguyen, Nguyen and Nguyen, 2018a; Fang *et al.*, 2019). Furthermore, because the data in emails are unstructured, deep learning algorithms are the best choice for this work (Halgaš, Agrafiotis and Nurse, 2020). Moreover, using deep learning algorithms in phishing email detection proved its effectiveness, such as the result described in studies of (Deng, 2014; Nguyen, Nguyen and Nguyen, 2018a; Fang *et al.*, 2019; Halgaš, Agrafiotis and Nurse, 2020).

A GCN is a deep learning algorithm described as a type of convolutional neural network (CNN) that operates directly on graphs using structured data through semi-supervised learning to classify nodes. The GCN algorithm is commonly used in classification problems such as document classification; it simply converts the document classification problem into a node classification problem. The idea behind the GCN, as explained by (Kipf and Welling, 2017; Yao, Mao and Luo., 2019), is to first construct the corpus graph by building a single large graph from an entire email corpus, so that email words and the email itself are represented as nodes. The graph is then fed into a convolutional network.

Moreover, using NLP on the email body text to classify email body features in phishing detection has proven to be successful, as shown in the studies of (Bergholz *et al.*, 2008; Toolan and Carthy, 2010; Egozi and Verma, 2018; Peng, Harris and Sawa, 2018a; Fang *et al.*, 2019). Deep learning outperformed several NLP tasks to achieve state-of-the-art results in tasks such as text classification (Glorot, Bordes and Bengio, 2011; Lai *et al.*, 2015; Prasad and Reddy, 2019), machine translation, and information classification (Nguyen and Grishman, 2015a, 2015b, 2016). Furthermore, deep learning algorithms contribute to improving phishing detection systems based on NLP (Egozi and Verma, 2018; Nguyen, Nguyen and Nguyen, 2018a; Fang *et al.*, 2019).

Lai et al. (2015) proposed a text classification model based on the recurrent convolutional neural network (RCNN.(They used a recurrent structure to capture the contextual information of the text. They also used a CNN to construct a representation of the text. They applied the model to four datasets and compared the accuracy to a CNN model, a recurrent neural network (RNN) model, and other conventional models. They found that the RCNN model outperformed all other models tested.

Yao et al. (2019), proposed a novel method for text classification based on graph neural networks. The main idea was to represent an entire corpus as a heterogeneous graph and learn word and document embeddings mutually using graph neural networks. They applied the model to four text-size datasets and compared the results with multiple state-of-the-art text classification and embedding methods. This model proved to be effective with a 97% accuracy rate.

Fang et al. (2019), produced the THEMIS classification model, which works on the email body and header. Instead of using feature extraction for text, the authors used deep learning. They used the word2vec tool to represent emails, and they extracted the char-level email header, word-level email header, char-level email body, and word-level email body. The RCNN deep learning algorithm was used to develop the model. The accuracy of the THEMIS model was promising at 99%, proving the efficacy of using NLP for email phishing detection.

Egozi and Verma (2018) proposed a classification model based on NLP and machine learning, with a focus on novel fea-

tures. The model used 26 features, four of which were novel. These features were applied across 17 machine learning models, and they produced a true positive rate of 83% and a true negative rate of 96%.

Peng et al. (2018a) proposed a phishing detection model using NLP techniques. Their model tested the email's body sentences to detect the probability of phishing based on four malicious rules: command or malicious question, insistent tone, generic greeting, and malicious URL. An email was directly marked as a phishing attempt if the sentence contained a malicious URL or if the email violated two malicious rules. To generate a malicious question blacklist, they used a machine learning algorithm. The precision was 95%, and the recall was 91%.

Nguyen et al. (2018a) proposed a deep learning model that focused on using NLP at the word level and sentence level separately, then combined the output with an attention mechanism. The model precision was 99%, and the recall was 99.2%.

Almomani et al. (2013) proposed a phishing dynamic evolving neural fuzzy framework, which adjusts the evolving connectionist system. It dynamically detected phishing emails, including unknown zero-day phishing emails, before it reached the recipient's mail. Their framework was assumed to learn continuously with a minimal memory footprint and reduced the complexity with minimum rules for email classification.

Halgaš et al. (2020) proposed a classifier using an RNN to classify emails as legitimate or phishing based on the email's text. The classifier proved its effectiveness and accuracy. Moreover, it could be used as a complement to the existing classifiers.

This study uses the NLP of email body features and deep learning algorithms using GCN to build an effective phishing email detection classifier to increase the probability of automatically detecting the possibility that an email is a phishing email.

3. The proposed phishing detection model

To build the proposed classifier, we go through three main phases. First, data from phishing and legitimate emails are collected and prepared for training and testing, as described in Section 3.1. The proposed detection model is constructed through deep learning GCN algorithms, as described in Section 3.3. Training data are fed to the classifier to construct the model and prepare it for evaluation. Then, the final stage is to test the proposed classifier in a supervised approach using the testing data to validate the model. A diagram describing the workflows of the research procedure is shown in Fig. 1.

Phishing email detection is a binary classification problem, where the classifier should distinguish between two kinds of Emails: phishing and legitimate. For a classification phishing task, the input is a collection of instances (emails). Each instance, also known as a record, is represented by a tuple (x, y), where x is the attribute set and Y is the class label or target attribute (Baxter et al., 2011). It is a supervised learning task as there is prior knowledge of the output targets set. The goal of a classification task is to predict the class of an unlabelled sample. If we have a group of finite class C= {c1, c2,c3,.....} , an unlabelled sample x is mapped to one class in C. For the case of phishing email detection, the classification is a representation of an email with the label set {phish, legitimate}.

In greater detail, the classification process starts with feeding the classifier with emails that are processed to extract the body text. They are cleaned of noise and irrelevant information, tokenised, and prepared for graph construction. The GCN is trained to classify phishing and ham emails. The prosed detection classifier model used in this research is illustrated in Fig. 2, and the description of each step is provided in the following subsection.

3.1. *Data pre-processing*

The data need to be cleaned to exclude needless characters and words. After the data have been cleaned, formal feature extraction techniques can be used. Therefore, the preparation of email data involves:

1 Extracting the body text
2 Eliminating white spaces caused by text parsing
3 Lowercasing all characters and removing nonalphanumeric characters

This study applied various pre-processing techniques to the emails, including email cleaning, stop- and rare-word filtering, and tokenization. Each is described as follows:

3.1.1. *Email cleaning*

This process cleans emails of irrelevant information and non-English characters. Nonalphanumeric characters and nonspecial characters, such as "?", "!", and "'", are removed using Python Regex. White spaces are also removed. Fig. 3 shows the pseudocode used and an example.

3.1.2. *Tokenization*

The tokenization process breaks each email into words based on white spaces. These words are referred to as tokens. In this study, we use a split() function. Fig. 4 shows an example of the tokenization process.

3.1.3. *Stop-word and rare-word removal*

Stop words are very common words; they are usually words that help build ideas but do not carry any significance themselves, such as conjunctions, articles, prepositions, and so on. We used the stop words list from the Natural Language Toolkit (NLTK). The stop words list contains words like "off", "no", "aren", "too", "an", "being", "only", "ll", "o", "its", "them", and "mightn". Moreover, any word that appears less than seven times is removed. The pseudocode of the stop-word and rare-word removal process is shown in Fig. 5.

3.2. *Graph construction*

After cleansing the dataset, the next step is to construct a single large graph from the entire email corpus, with the words and email used as nodes. The edges that connect the word nodes depend on the co-occurrence information between two

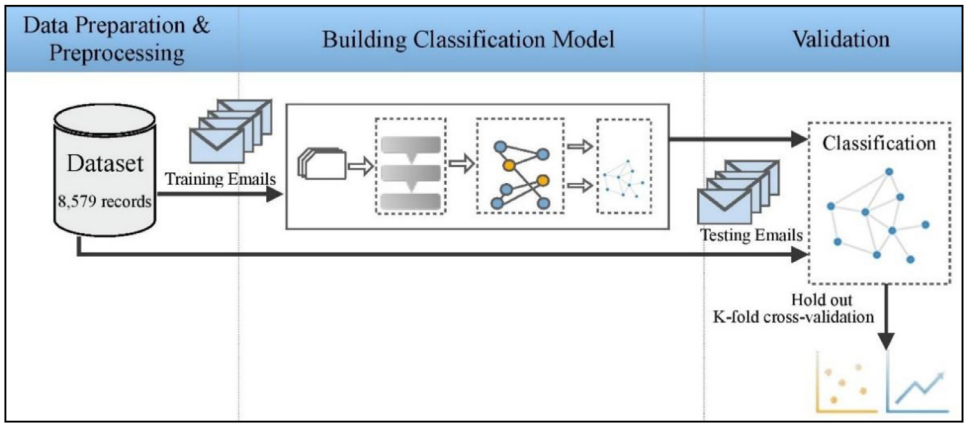


Fig. 1 – The main phases of research workflow.

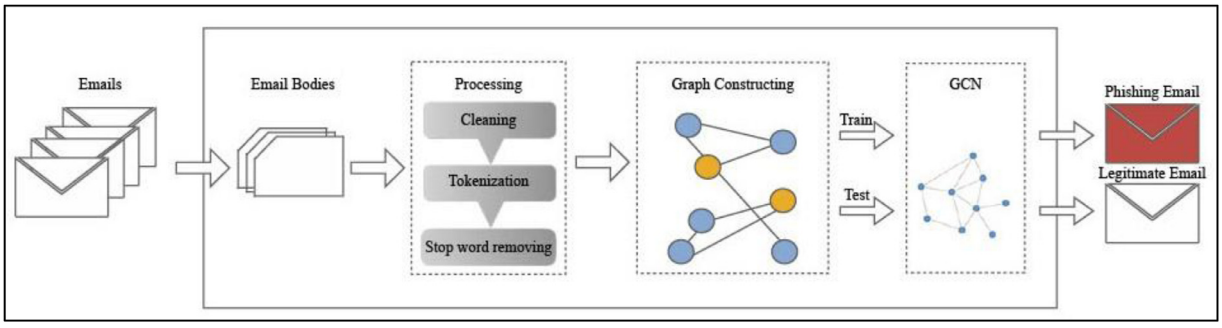


Fig. 2 – The phishing Email detection classifier model.

Function

```
string = re.sub(r"[^A-Za-z0-9(),!?\'\`]",
" ", string)
```

Example

Input	**** I will meet you at 9 :) *****
Output	I will meet you at 9

Fig. 3 – Email cleansing.

Input	I like python
Output	['i','like','python']

Fig. 4 – Email tokenization.

```
For each word in words:

  IF word not in stop_words and
word_freq[word] >= 7:

Email_words.append(word)
```

Fig. 5 – stop word removal.

words. The edges between a word and an email are constructed using the word frequency and the word's email frequency. Then, the text classification problem becomes a node classification problem (Kipf and Welling, 2017; Yao, Mao and Luo., 2019).

As described in (Kipf and Welling, 2017; Yao, Mao and Luo., 2019), we denote the graph as G, where G is an undirected graph and G = (V, E), V is the collection of nodes N, and E is the collection of the edges. The N nodes are the words and the emails. To construct the graph, an adjacency matrix A is introduced where $A\hat{I}R^{N\times N}$, and A contains every word and every email in the corpus as rows and columns to record the correlation between them, along with the number of co-occurrences of each word and another word together on the corpus. The values in the A matrix are the weights for each relation. The degree matrix of A, denoted as D, contains the correlation between each node and the other nodes in a diagonal matrix, $D\hat{I}R^{N\times N}$ and $D_{ii} = \sum_i A_{ij}$. X is a feature vector matrix where X is the identity matrix of A, which shows each word and email in a one-hot vector. Fig. 6 shows the representation of the GCN graph concept using a section of the corpus.

As described in (Kipf and Welling, 2017; Yao, Mao and Luo., 2019), A_{IJ} shows the edge between nodes *i* and *j*. The value for edges is calculated as follows: if *i* and *j* are the same node, then the value is equal to 1. If *i* and *j* are different words, the value is their co-occurrence. The term frequency–inverse document

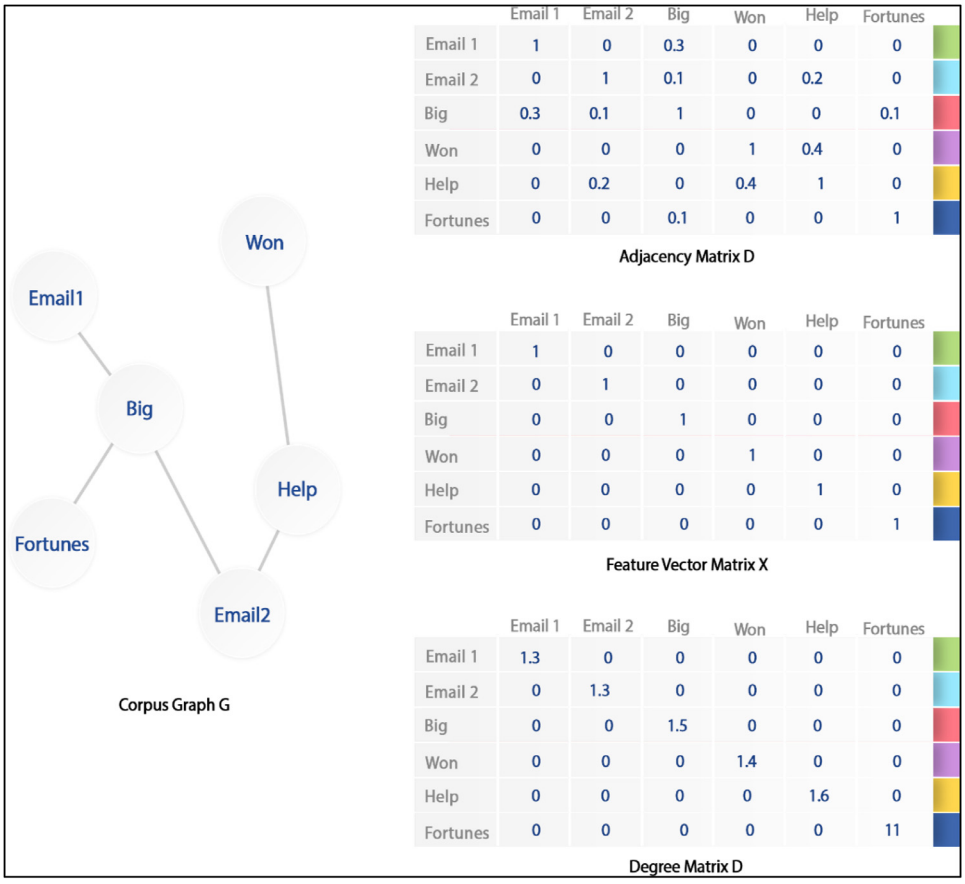


Fig. 6 – Concept of GCN graph representation.

frequency (TF–IDF) is used to calculate the edge between a word denoted using *i* and an email denoted by *j*.

The TF–IDF is commonly used in document search applications (Yao, Mao and Luo., 2019). It represents the correlation between a word and a specific email. It is calculated by dividing the number of times a word appears in an email by the total number of all emails that contain these words. Therefore, the more emails that contain the word, the lower the TF–IDF value for the word because there is no significant correlation between the email and the word, as it appears in many other emails.

As mentioned in (Yao, Mao and Luo., 2019), for the words co-occurrence, point-wise mutual information (PMI) is used. It calculates the word associations as shown in the following equation:

$$PMI(I,J) = \frac{P(I,J)}{P(I)P(J)} \tag{1}$$

Where P(I,J) is probability of adjacency for word I and word J, P(I) and P(J) are the probability of word I, J in the corpus respectively.

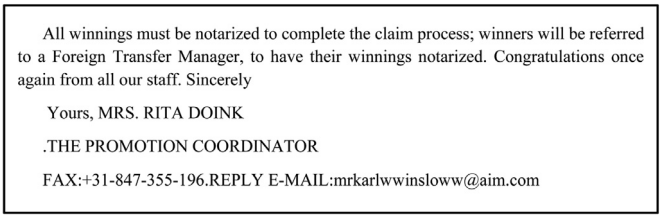
3.3. *The GCN classifier*

Each node in the graph takes the information of all node neighbours, including itself. Then, the calculated value is sent to a neural network (Yao et al., 2019). Node vectors are scaled or normalised before passing the neighbours' information to a specific node. Scaling is accomplished by multiplying A by its degree matrix D. The vectors are scaled because nodes with many neighbours have large vectors v, whereas nodes with few neighbours have small aggregate vectors, which can lead to vanishing gradients and prevent the neural network from further training. Thus, we scale these vectors to overcome the potential issues. A is multiplied by its D inverse two times to scale both rows and columns, and it contains the information of all nodes' neighbours, including itself.

Consequently, the A matrix will be scaled to be:

$$\tilde{A} = D^{-0.5}AD^{-.5} \tag{2}$$

In the GCN, when we use one convolutional network layer, the node can only capture information from its nearest neighbours. Therefore, the more convolutional layers are used, the more information from neighbours is integrated. Thus, the more information is needed about distant neighbours, the more layers we should have. The task of gathering information about each node from its neighbours proceeds in parallel for each node. With two layers, we repeat the information-gathering task of the nearest nodes only. Based on (Kipf and Welling, 2017; Yao, Mao and Luo., 2019), we typically need two to three layers in the graph.


All winnings must be notarized to complete the claim process; winners will be referred to a Foreign Transfer Manager, to have their winnings notarized. Congratulations once again from all our staff. Sincerely

Yours, MRS. RITA DOINK

.THE PROMOTION COORDINATOR

FAX:+31-847-355-196.REPLY E-MAIL:mrkarlwwinsloww@aim.com

Fig. 7 – Example of a phishing e-mail.

Based on (Yao, Mao and Luo., 2019), the feature matrix of the first layer calculated as

$$L^{(1)} = P\left(\tilde{A}\, X\, W_0\right) \tag{3}$$

$\tilde{A}$ is the scaled adjacency matrix that is previously calculated as

$$\tilde{A} = D^{-0.5} A D^{-.5} \tag{4}$$

X is the identity matrix of A, W is the weight matrix that can be computed using gradient descent, and p is the activation function, such as sigmoid or ReLU. The feature matrix of the rest of the layers could be calculated as

$$L^{(i+1)} = P\left(\tilde{A}\; L^{I}\, W_0\right) \tag{5}$$

The email feature matrix is provided as an input to the neural network, which is composed of an input layer, hidden layers, and an output layer. The input layer accepts external data into the model, which are email features in this study, and the hidden layers simultaneously learn all inspections in the training set to produce the output. The output layer provides a classification of 1 for a phishing email or 0 for a legitimate email.

4. Experiment and evaluation

The computer used for this experiment was a personal computer with the following specifications: 2.3 GHz Quad-Core Intel Core i7 processer, 32 GB 3733 MHz LPDDR4X RAM, and a 64-bit Linux operating system. In addition, the Python programming language was used.

4.1. *Dataset*

We have used in this study a publicly available dataset, which is the fraud dataset (Radev, 2008). The dataset is balanced and labeled to be used in supervised learning. The phishing and legitimate Emails are labeled as (1) and (0), respectively. The record contains only the Email's body in the English language. The downloaded file was in CSV format then converted to a text file. The used dataset consists of:

- 3685 phishing Emails, Fig. 7 illustrates an example of phishing Emails.
- 4894 legitimate Emails.

Table 1 – The classification confusion matrix.

Actual	Predict 0 - Legitimate (ham)	1 - phishing
0 - Legitimate (ham)	TN	FP
1 - phishing	FN	TP

4.2. *The GCN classifier setting*

Neural networks algorithms have various parameters to set, such as number of GCN layers, maximum number of iterations, early stopping of window size and so on. Parameters are set to achieve maximum accuracy based on work of (Kipf and Welling, 2017) to be as (learning rate: 0.1; Solver: Adam; Shuffle = True; Early stopping: After 10 epochs; Number of GCN layers: 2 layers; Loss function: cross-entropy error).

4.3. *Evaluation and results*

To evaluate the effectiveness of the phishing classifier, the initial dataset, a prelabelled set of instances, was split into two disjoint sets: the training and testing sets (Hlaváč, 2016). First, the training set was used to allow the neural network to learn how to classify emails. Second, the testing set held the data to be fed to the neural networks without labelling.

Then, a neural network was used to predict the classes. The testing set was used to evaluate the performance of the neural network predictions by comparing the predictions to the manually labelled decisions. Following the evaluation metrics used in the literature of (Kipf and Welling, 2017; Gupta, Arachchilage and Psannis, 2018; Nguyen, Nguyen and Nguyen, 2018b; Peng, Harris and Sawa, 2018b; Yao, Mao and Luo., 2019; IC3, 2020), we used accuracy, precision, and recall to evaluate the model.

False positive (FP) is the number of legitimate (ham) emails misclassified as phishing, true negative (TN) is the number of legitimate (ham) emails classified as legitimate (ham), false negative (FN) is the number of phishing emails misclassified as legitimate (ham), and true positive (TP) is the number of phishing emails classified as phishing emails. The classification confusion matrix is shown in Table 1.

To evaluate the models, the following metrics are used.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{6}$$

Table 2 – 3-fold cross-validation results.

Fold	iteration	Precision	Recall	F1-score	Avg precision	Avg recall	Avg f1-score	Accuracy
1st	0	0.982	0.991	0.986	0.9867	0.9868	0.9867	0.9868
Fold	1	0.991	0.982	0.986				
2nd	0	0.990	0.988	0.989	0.9911	0.9900	0.9909	0.9913
Fold	1	0.992	0.993	0.9934				
3rd	0	0.969	0.992	0.980	0.9820	0.9855	0.9836	0.9841
Fold	1	0.994	0.9781	0.986				

Table 3 – The classification confusion matrix.

Actual	Predict 0 - Legitimate (ham)	1 - phishing
0 - Legitimate (ham)	4796	98
1 - phishing	74	3611

Table 4 – The accuracy metrics values.

Accuracy	98.2%
Precision	98.5%
Recall	98.3%
f-measure	98.5%

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{7}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{8}$$

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN} -} \tag{9}$$

There are two methods that are commonly used to evaluate a classifier's effectiveness: the holdout method and the k-fold cross-validation method (Rithchie, 2018). In a holdout split, the classifier is evaluated by calculating the error rate on a new data set, which is the testing set rather than the training set. Practically, it is common to use two-thirds of the dataset for training and the rest of the data for testing. In the k-fold cross-validation method, the training set is randomly divided into k disjoint sets of equal size, where each part has roughly the same class distribution. The classifier is trained k times, each time with a different set held out as the testing set. The estimated error is the mean of these k errors.

For the evaluation, both methods were used. Hold out split depends on splitting the data into two datasets, one for training and the other for testing. We used 70% of data for training (6,005 rows) while 30% for testing (2574 rows). Moreover, we used 3-folds cross-validation method, by dividing the dataset into three groups. The classifier is trained three times, each time with a different set held out as a test set. The result of the 3-folds cross-validation presented in Table 2.

The results of confusion matrix, explained previously in Table 2, is presented in Table 3. According to Fang et al.Fang *et al.*, 2019), it is expected that the values of TP and TN is large, whereas the values of FP and FN is small. The results confirm what is expected and can be used to obtain more evaluation metrics. Based on Eq. (9), the measured FPR is 0.015, indicating a low probability of classifying legitimate Emails as phishing.

Evaluation of the classifier is calculated using accuracy, precision and recall metrics as described in Section 4.3 in Eqs. (6–(8). Table 4 shows the result.

The measures of recall, accuracy, precision, and f-measure are compared with other published studies to evaluate the effectiveness of our model. The results are presented in Table 5. It shows that the GCN classifier competes with other machine learning classifiers in phishing email detection with a high accuracy rate. It is good to note that Bergholz et al. (2008) who achieved higher accuracy, used machine learning techniques depending on feature extractions that required a domain expert to intervene in the process to reduce the complexity of the data and make patterns more visible to learning algorithms. However, our model eliminates the need for a domain expert and handcrafted feature extraction. Furthermore, they used eight link features that will not work with emails that do not contain links. In addition, Fang et al. (Fang *et al.*, 2019) used the header and body to detect phishing emails, while our model uses only the body text, which reduces the memory complexity. Moreover, we have better precision and recall than (Nguyen, Nguyen and Nguyen, 2018a) and short learning and classification time.

In comparison to other deep learning methods that use images, this model is more efficient as involving the processing of images could lead to an increase in the run time and space as shown in the work of (Bergholz *et al.*, 2010) who used model-based features including images. They also found that even in emails with photos, the message text and information about links were more effective for phishing detection.

Theoretically, the memory requirement for the GCN algorithm is O(E) and the time complexity is O(LED + LND2). Where L is the number of layers, N is the total number of nodes and D is feature dimension and E is the number of the graph's edges (Wu et al., 2021). Experiments have shown that running time of the model is on average of 3.89 minutes for training 6005 emails and Average of 1.08 sec for testing 2574 emails. which is considered short time in comparison

Table 5 – Comparison of related works in term of the accuracy, precision, recall and f-measure.

Reference	Technique	accuracy	precision	recall	f-measure
(Lai *et al.*, 2015)	Text classification for phishing detection based on RCNN.	96.94%	-	-	-
(Nguyen, Nguyen and Nguyen, 2018a)	Deep learning hierarchical long short-term memory networks (H-LSTMs	99%	97%	95%	96%
(Halgaš, Agrafiotis and Nurse, 2020)	Deep learning	96.74%	97.45%	95.98%	96.71%
(Peng, Harris and Sawa, 2018a)	Machine learning algorithm and NLP	-	95%	91%	-
(Bergholz *et al.*, 2008)	Machine learning algorithm using semantic features	99.88%	100%	98.93%	99.46%
(Fang *et al.*, 2019)	Deep learning and NLP	99.84%	99.66%	99.00%	99.33%
Our model	GCN	98.2%	98.5%	98.3%	98.55%

to (Nguyen, Nguyen and Nguyen, 2018a) who took 60:30 to 180:07 minutes for less than 6000 emails.

5. Conclusion and future work

Phishing email is one of the fastest-growing cybercrimes on the Internet, and it affects businesses and individuals alike. It also costs billions of dollars every year. Phishing methods are rapidly changing and depend on the intelligence of the hacker. Therefore, taking advantage of machine deep learning to compete and match human intelligence is a perfect solution.

The main aim of this study is to propose a solution that improves the performance and accuracy of phishing email classification and detection. It examined a phishing email detection classification model using NLP and a deep learning algorithm based on GCN. The model was tested with a supervised learning approach and was proved to improve classification accuracy and performance in comparison to other deep learning methods with a short time for training and a high true positive rate, true negative rate, hight recall, precision, f-measure, and overall accuracy.

It highlights the significance of examining the text features of an email body to detect phishing because the text features are a new research direction in the email phishing detection area, with only a few studies focusing on the email's body text. This research was among the first, to the best of our knowledge, to examine the use of the GCN algorithm in detecting phishing in the email body text. This will lead to more investigation in the field of phishing detection in text, whether in emails, social media messages, or even phishing websites. Furthermore, we plan to conduct a future study using a non-English corpus. In addition, because the solution's concept depends on text classification, we assume it can be applied to email categorisation tasks.

Moreover, the proposed detection classifier can accurately detect phishing emails based on body text that has not been seen before, so we assume its effectiveness in detecting zero-day phishing attacks. Hence, further investigation will be done to evaluate its effectiveness in detecting zero-day phishing emails that have been spreading widely recently.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Areej Alhogail: Supervision, Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Visualization. **Afrah Alsabih:** Software, Validation, Investigation.

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